

Moral numbers - comparison between humans and LLMs

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Abstract

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1 Comparisons between humans and LLMs

1.1 Function and definitions etc copied from main file

1.2 Other definitions and settings

1.3 Methods

We compared the responses of humans to those of several frontier (i.e., state-of-the-art) large language models (LLMs). We focused on three experiments: In Experiment 2, we asked whether LLMs are sensitive to the beneficiary:victim Weber ratio in moral and economic scenarios. In Experiment 3, we directly pitted Weber ratios against harm (the number of victims in Experiment 3a) and utility (the net number of survivors in Experiment 3b). Finally, in Experiment 6, we tested how LLMs react to manipulations of the vividness of the victims’ deaths.

1.3.1 Models and parameters

We tested GPT-4o mini, GPT-4.1, GPT-4.1 mini, GPT-5 mini (minimal reasoning), GPT-5 mini (medium reasoning), GPT-5 mini (high reasoning), Claude Sonnet 4.5, Gemini 2.5 flash, Gemini 2.5 pro, Mistral Medium, Deepseek, and Qwen-plus. We chose these models because they reflect a range of jurisdictions (China, France and USA) and thus potentially differences in training corpora and alignment procedures (i.e., post-training procedures used to shape model behavior according to values determined by the model creators and their jurisdictions). We used different versions of OpenAI’s GPT models because, during initial non-systematic exploration, some models seemed more deontological than others, and because this allowed us to set the reasoning effort parameter in GPT-5 mini. (This parameter controls how many internal and intermediate reasoning steps the model generates before producing its final response.) We also explored several smaller models run on a local computer through ollama (<https://ollama.com/>), namely Llama 3.2 (3.2B), Mistral (7.2B), and Mistral nemo (12.2B)

For models that support these options, we used the parameters given in Table 1 or their equivalents. The system prompts are given in Table 2.

Table 1: API parameters and the models for which each was included in the request body. GPT-4 = GPT-4o mini, GPT-4.1, GPT-4.1 mini; GPT-5 = GPT-5 mini (all reasoning effort levels); Local = Llama 3.2, Mistral 7.2B, Mistral nemo (Ollama). Dashes indicate parameters defined in LLM_OPTIONS but not passed to any model. `max_output_tokens` is passed under model-specific field names (`max_tokens` for GPT-4, Claude, Mistral, Qwen; `maxTokens` for Deepseek).

Parameter	Value	Description	Available in
<code>max_attempts</code>	10.0	Max. API call retries on failure	All models (client-side)
<code>max_output_tokens</code>	10.0	Max. tokens in model response	GPT-4, Claude, Mistral, Deepseek, Qwen
<code>temperature</code>	0.1	Response randomness (0 = deterministic)	GPT-4, Claude, Gemini, Mistral, Deepseek, Qwen, Local
<code>top_k</code>	40.0	Top-K token sampling cutoff	Qwen
<code>top_p</code>	1.0	Nucleus sampling probability	Gemini
<code>n</code>	1.0	Number of completions per request	GPT-4, GPT-5, Mistral, Qwen
<code>frequency_penalty</code>	0.0	Penalizes already-frequent tokens	—
<code>presence_penalty</code>	0.0	Penalizes tokens seen so far	—
<code>num_ctx</code>	8192.0	Context window size (tokens)	Local

1.3.2 Procedure

As participants completed multiple trials with different scenarios in each experiment, we treated the LLMs in the same way. Specifically, we treated the trials as an ongoing incremental conversation. On the first trial, the API calls contained the system prompt as well as the first scenario. On the second trial, they contained the system prompt, the first scenario together with the model’s answer on the first trial (provided as a message from the *assistant* role or the equivalent in other models), as well as the second scenario. Finally, on the last trial, the API calls contained the system prompt, all previous scenarios together with the model’s answers, as well as the last scenario.

Our experiments used a Latin square design. We obtained 10 sets of independent responses per cell in the Latin square, treating each response as a simulated participant. In Experiment 6, these cells were crossed with two different sets of numbers of victims and beneficiaries. The resulting numbers of cells, (simulated) participants, trials per participant, and responses are given in Table 3, with a total of 1160 simulated participants and a total of 13520 responses.

1.3.3 Initial verification

To validate the LLM results, we recorded the number of valid responses for each experiment and condition, where a valid response is a numerical score with no other information. As shown in Table 4, when excluding Claude Sonnet 4.5, data loss was at most 1.5%. In the economic conditions of Experiments 3a and 3b, Claude Sonnet 4.5 returned many undefined responses or responses with extraneous content beyond the requested numerical values. As this pattern replicated in independent simulations, it appears that Claude Sonnet 4.5 has difficulties following instructions in the economic conditions in Experiments 3a and 3b, but showed similarly low rates of data loss as the other models in the other experiments.

Table 2: System prompts used in all experiments, given the LLMs general instructions on how to behave.

Rating on 6-point scale (all experiments but Exp. 3) You are an average adult with no formal training in philosophy, ethics, or law. Your sense of right and wrong comes from everyday life—social norms, gut feelings, personal experience. You don’t try to be logically consistent or follow abstract moral rules. You react like a regular person would: practically, emotionally, and socially. You speak plainly, without academic or philosophical jargon. You can hesitate, contradict yourself, or be unsure—just like people often are. Please give your gut-level reaction, even if you’re not sure. When asked about the acceptability of an action, respond on a 6-point scale (1 = completely unacceptable, 6 = completely acceptable). Respond **only** with a single digit (1–6). Do **not** add words, punctuation, or explanations. Any other output but the number will make your response unusable. Choice between two situations (Exp. 3) You are an average adult with no formal training in philosophy, ethics, or law. Your sense of right and wrong comes from everyday life—social norms, gut feelings, personal experience. You don’t try to be logically consistent or follow abstract moral rules. You react like a regular person would: practically, emotionally, and socially. You speak plainly, without academic or philosophical jargon. You can hesitate, contradict yourself, or be unsure—just like people often are. Please give your gut-level reaction, even if you’re not sure. When asked about the relative acceptability of two actions, respond on a 6-point scale (1 = Option 1 is much more unacceptable, 6 = Option 2 is much more acceptable). Respond **only** with a single digit (1–6). Do **not** add words, punctuation, or explanations. Any other output but the number will make your response unusable.
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Table 3: Experimental design summary for LLM simulations. Each cell in the Latin square design was tested with 10 independent response sets, with each response treated as a simulated participant. The table shows the number of conditions (cells), simulated participants, trials per participant, and total responses for each experiment. In Experiment 6, *Group* refers to two different sets of victim and beneficiary numbers crossed with the neutral/vivid manipulation, resulting in four conditions total.

		Number of			
condition	Group	cells	participants	trials/participant	responses
Exp. 2					
Moral		10	100	10	1000
Economic		10	100	10	1000
Exp. 3a					
Moral		12	120	12	1440
Economic		12	120	12	1440
Exp. 3b					
Moral		12	120	12	1440
Economic		12	120	12	1440
Exp. 6					
Neutral	Group 1	12	120	12	1440
Neutral	Group 2	12	120	12	1440
Vivid	Group 1	12	120	12	1440
Vivid	Group 2	12	120	12	1440

Table 4: Valid response rates across experiments and models. Valid responses were numerical scores (1-6) with no additional text. Maximum data loss (excluding Claude Sonnet 4.5) was 1.5%. Claude Sonnet 4.5 showed substantial data loss in economic conditions of Experiments 3a and 3b, returning undefined responses or extraneous content beyond the requested numerical values. This pattern replicated across independent simulations, suggesting systematic difficulties following instructions in these specific conditions.

Experiment	Condition	Expected value	Most frequent value	Min	Max	Max data loss (%)
All models						
Exp. 2	Moral	1000	1000	1000	1000	0.000
Exp. 2	Economic	1000	1000	1000	1000	0.000
Exp. 3a	Moral	1440	1440	1320	1440	8.333
Exp. 3a	Economic	1440	1440	237	1440	83.542
Exp. 3b	Moral	1440	1440	1320	1440	8.333
Exp. 3b	Economic	1440	1440	178	1440	87.639
Exp. 6	Neutral	2880	2880	2880	2880	0.000
Exp. 6	Vivid	2880	2880	2873	2880	0.243
All models excluding Claude Sonnet 4.5						
Exp. 2	Moral	1000	1000	1000	1000	0.000
Exp. 2	Economic	1000	1000	1000	1000	0.000
Exp. 3a	Moral	1440	1440	1439	1440	0.069
Exp. 3a	Economic	1440	1440	1418	1440	1.528
Exp. 3b	Moral	1440	1440	1435	1440	0.347
Exp. 3b	Economic	1440	1440	1440	1440	0.000
Exp. 6	Neutral	2880	2880	2880	2880	0.000
Exp. 6	Vivid	2880	2880	2873	2880	0.243

1.4 Results

1.4.1 Experiment 2

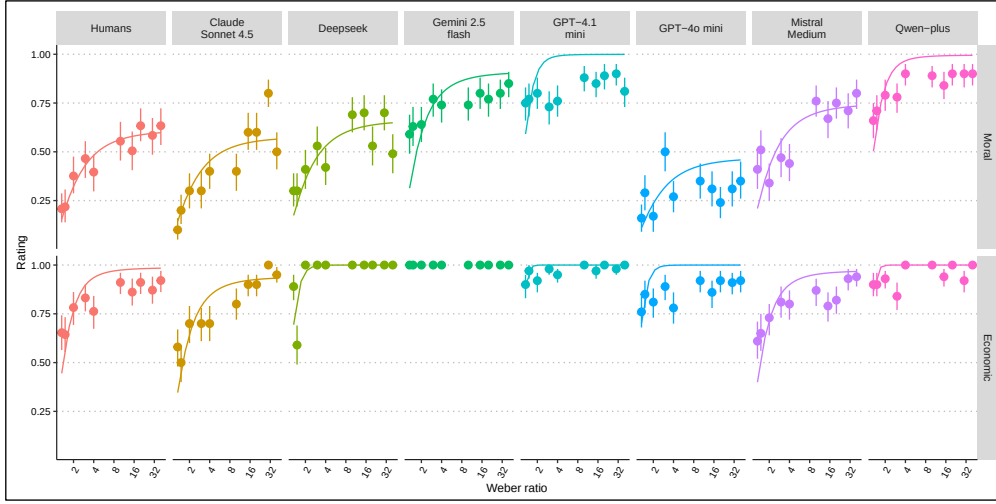


Figure 1: Acceptability of sacrificial decisions as a function of beneficiary:victim Weber ratios in Experiment 2 for frontier models. Data show *binarized responses* (acceptable/unacceptable) for moral (top) and economic (bottom) scenarios. Points represent mean acceptability with 95% confidence intervals. Curves show psychophysical model fits for the average responses across simulated participants. All models showed sensitivity to Weber ratios in moral scenarios, though some models (GPT-4.1 mini, GPT-5 mini variants, Qwen-plus) were substantially more lenient than humans.

1.4.1.1 Frontier models In Experiment 2, we asked whether the models would be sensitive to the beneficiary:victim Weber ratio in moral and economic scenarios. The results for frontier models and the binarized ratings (responses transformed acceptable/unacceptable decisions on which our model is based) are shown in Figure 1. For moral scenarios, all LLMs showed some sensitivity to Weber ratios, though the quality of fit to our psychophysical model varied. However, some models were much more likely to endorse a sacrificial decision than humans, namely GPT-4.1 mini, GPT-5 mini (with all reasoning effort settings) and Qwen-plus.

In contrast, for economic scenarios Gemini-2.5 flash and GPT-5 mini (all reasoning effort settings) showed no sensitivity to Weber ratios when considering binarized responses. Nevertheless, Figure 2 shows that, when considering the raw (rather than binarized) responses, these models also showed some limited sensitivity to Weber ratios.

Model	Condition	w	
		M	SE
Humans			
Humans	Moral	1.401	0.064
Humans	Economic	0.426	0.061
	Moral	0.138	0.010
	Economic		
Frontier models			
Claude Sonnet 4.5	Moral	1.518	0.147
Claude Sonnet 4.5	Economic	0.577	0.062
Deepseek	Moral	1.240	0.106
Deepseek	Economic	0.229	0.040
Gemini 2.5 flash	Moral	0.647	0.085

(continued)

Model	Condition	M	SE
Gemini 2.5 flash	Economic		
GPT-4.1 mini	Moral	0.292	0.072
GPT-4.1 mini	Economic	0.135	0.014
GPT-4o mini	Moral	2.000	0.290
GPT-4o mini	Economic	0.225	0.047
Mistral Medium	Moral	1.014	0.108
Mistral Medium	Economic	0.486	0.065
Qwen-plus	Moral	0.364	0.057
Qwen-plus	Economic	0.149	0.026
Local models			
Llama 3.2 (3.2B)	Moral	0.757	0.141
Llama 3.2 (3.2B)	Economic	0.818	0.186
Mistral (7.2B)	Moral	2.000	0.374
Mistral (7.2B)	Economic	0.827	0.222
Mistral nemo (12.2B)	Moral	2.000	0.724
Mistral nemo (12.2B)	Economic	1.698	0.667

Figures 1 and 2 also show that ratings were substantially higher for economic scenarios than for moral scenarios, mirroring human participants. Accordingly, and as shown in Table ??, the w parameter was higher in the moral than in the economic condition for most models, with the exception of Llama 3.2 (3.2B), where the two values were comparable.

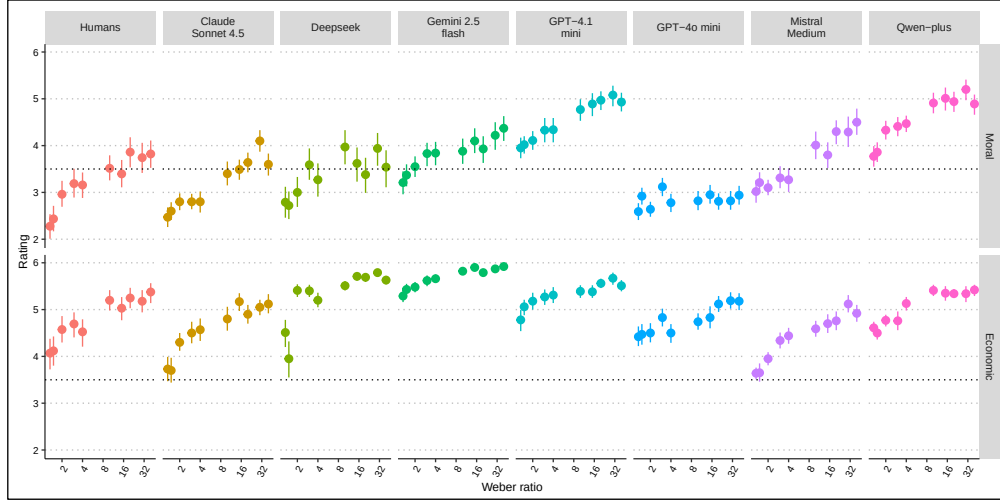


Figure 2: Acceptability ratings (1-6 scale) as a function of beneficiary:victim Weber ratios in Experiment 2 for *frontier models*. Data show *raw ratings* for moral (top row) and economic (bottom row) scenarios. Points represent mean ratings with 95% confidence intervals. Curves show psychophysical model fits for the average responses across simulated participants. Dotted horizontal lines indicate the neutral response (3.5).

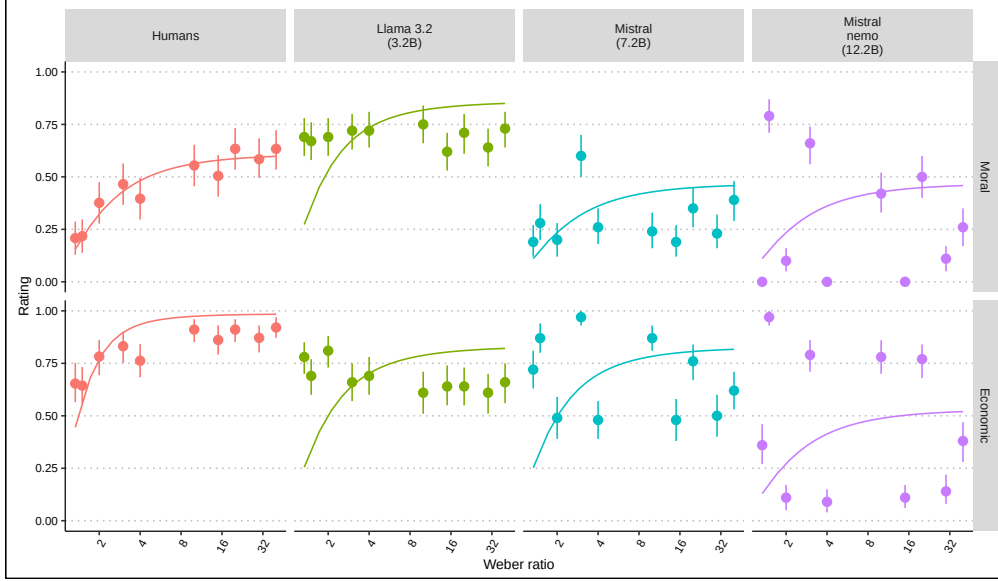


Figure 3: Acceptability of sacrificial decisions as a function of beneficiary:victim Weber ratios in Experiment 2 for *local models*. Data show binarized responses for moral (top) and economic (bottom) scenarios. Points represent mean acceptability with 95% confidence intervals. Curves show psychophysical model fits for average responses across simulated participants. Unlike frontier models, local models showed no discernible sensitivity to Weber ratios in either moral or economic scenarios, though they did show higher acceptability rates for economic than moral decisions.

1.4.1.2 Local models In contrast to the frontier models, the (smaller) local models showed no discernible sensitivity to Weber ratios, neither in binary ratings (see Figure 3) nor in raw ratings (see Figure 4). However, they did show higher acceptability rates for economic than for moral decisions.

1.4.1.3 Summary Taken together, these results suggest that LLMs can show some sensitivity to Weber ratios in both moral and economic decisions, at least with massive frontier models though not with smaller local models. Further, the models rated economic sacrifices as more acceptable than moral sacrifices. However, the fit to our psychophysical model was of variable quality.

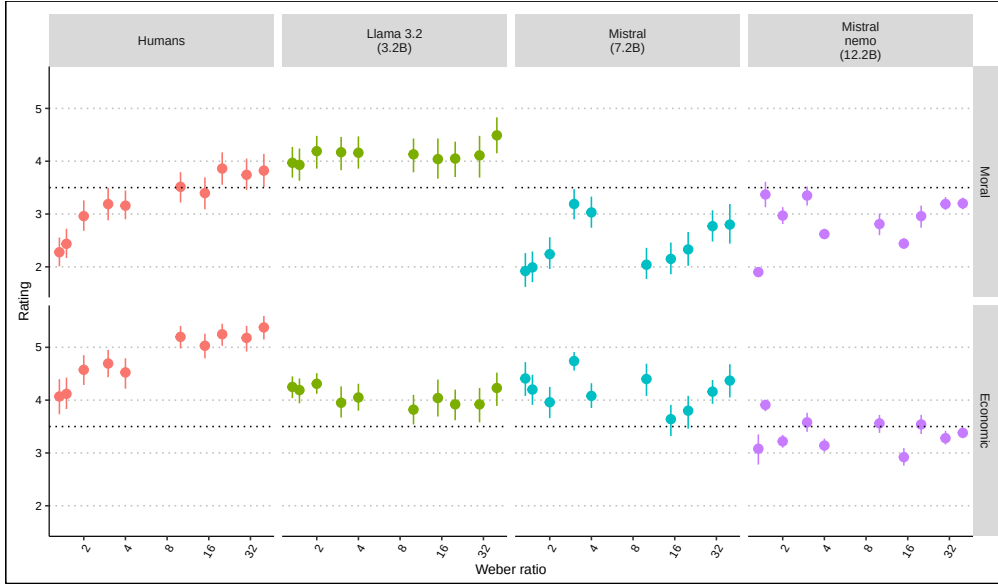


Figure 4: Acceptability ratings (1-6 scale) as a function of beneficiary:victim Weber ratios in Experiment 2 for local models. Data show *raw ratings* for moral (top row) and economic (bottom row) scenarios. Points represent mean ratings with 95% confidence intervals. Curves show psychophysical model fits. Dotted horizontal lines indicate the neutral response (3.5). Consistent with binary responses (Figure 3), local models showed no systematic sensitivity to Weber ratios.

1.4.2 Experiment 3

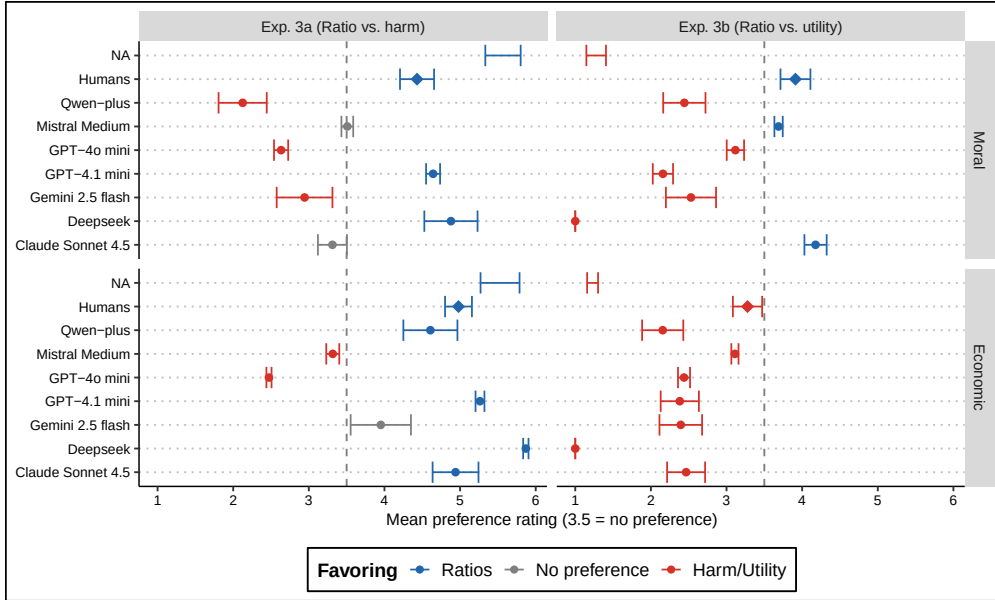


Figure 5: Forest plot of preferences between high-ratio and alternative options in Experiment 3 for frontier models. Points show mean preference ratings with 95% confidence intervals. The vertical dashed line at 3.5 indicates no preference; values above favour the high Weber ratio option, values below favour the alternative (low harm in Exp. 3a, high utility in Exp. 3b). Humans are shown as larger diamonds. Colour indicates the significantly favoured option (blue = ratio, red = alternative, grey = no significant preference).

1.4.2.1 Frontier models In Experiment 3, participants had to choose between two options. In Experiment 3a, one option had a higher beneficiary:victim Weber ratio and a higher number of victims (i.e., harm), while the other had a lower beneficiary:victim Weber ratio and also lower harm. In Experiment 3b, we similarly pitted Weber ratios against utility. In moral situations, humans prefer to maximize Weber ratios over minimizing harm or maximizing utility. In economic situations, in contrast, they still prioritize Weber ratios over harm, but slightly prioritize utility over Weber ratios.

The results from the LLM simulations are shown in Figure 5 and Table 7 (see Table 6 for detailed results). When pitting Weber ratios against harm in moral situations, four of the frontier models prioritized Weber ratios, four prioritized minimizing harm, while two had no preference. In economic situations, six of the LLMs prioritized Weber ratios, three prioritized minimizing harm, while one had no preference. Thus, only 4 out of 10 models reproduced the human pattern of results in moral situations, and 6 out of 10 in economic situations.

When pitting Weber ratios against utility, eight models prioritized utility in moral situations while two prioritized Weber ratios. In economic situations, all models prioritized utility. Both humans and LLMs thus prioritized utility in economic situations. In contrast, in moral situations, all but two models (Claude-Sonnet 4.5 and Mistral Medium) showed a qualitatively different behavior from humans, and prioritized utility over Weber ratio. However, neither Claude-Sonnet 4.5 nor Mistral Medium showed a preference for Weber ratio over harm, suggesting that they did not account for the human pattern of results either.

In other words, while some models showed a preference for Weber ratios over harm and some a preference for Weber ratios over utility, no model reproduced the human pattern of results.

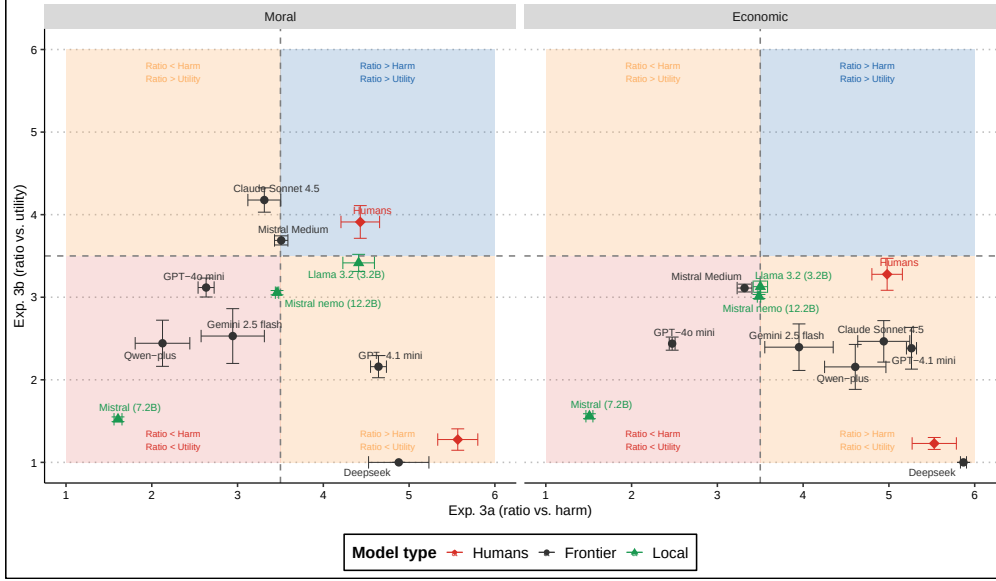


Figure 6: Joint preference ratings in Experiment 3 for each model and condition. The x-axis shows the mean preference rating in Experiment 3a (ratio vs. harm); the y-axis shows the mean rating in Experiment 3b (ratio vs. utility). Both axes use a 1–6 scale where 3.5 indicates no preference. Error bars show 95% CIs. Background colours indicate the quadrant: top-right (blue) = ratio preferred in both experiments, the human-like pattern. Colour and shape distinguish humans (red diamond), frontier models (dark circles), and local models (teal triangles). Quadrant labels are placed in the margins.

Table 6: Detailed results from Experiment 3. Mean ratings (M) with standard errors (SE) are shown on a 1-6 scale, where values greater than 3.5 indicate preference for the high Weber ratio option and values smaller 3.5 indicate preference for the alternative (low harm in Experiment 3a, high utility in Experiment 3b). The p-value tests whether the mean differs from 3.5 (no preference). The ‘favoring’ column indicates which option was significantly preferred.

model	condition	M	SE	p	favoring
Humans - Exp. 3a					
(Ratio vs. harm)					
Humans	Moral	5.57	0.119	0.000	Ratios
	Moral	4.43	0.115	0.000	Ratios
	Economic	5.53	0.132	0.000	Ratios
Humans	Economic	4.98	0.091	0.000	Ratios
Frontier models - Exp. 3a					
(Ratio vs. harm)					
Claude Sonnet 4.5	Moral	3.31	0.098	0.714	
Deepseek	Moral	4.88	0.179	0.000	Ratios
Gemini 2.5 flash	Moral	2.94	0.188	0.001	Harm
GPT-4.1 mini	Moral	4.64	0.047	0.000	Ratios
GPT-4o mini	Moral	2.63	0.048	0.000	Harm
Mistral Medium	Moral	3.51	0.040	0.836	
Qwen-plus	Moral	2.12	0.163	0.000	Harm
Claude Sonnet 4.5	Economic	4.94	0.155	0.000	Ratios
Deepseek	Economic	5.87	0.018	0.000	Ratios

Table 6: Detailed results from Experiment 3. Mean ratings (M) with standard errors (SE) are shown on a 1-6 scale, where values greater than 3.5 indicate preference for the high Weber ratio option and values smaller 3.5 indicate preference for the alternative (low harm in Experiment 3a, high utility in Experiment 3b). The p-value tests whether the mean differs from 3.5 (no preference). The 'favoring' column indicates which option was significantly preferred. (*continued*)

model	condition	M	SE	p	favoring
Gemini 2.5 flash	Economic	3.95	0.203	0.448	
GPT-4.1 mini	Economic	5.26	0.030	0.000	Ratios
GPT-4o mini	Economic	2.47	0.017	0.000	Harm
Mistral Medium	Economic	3.32	0.044	0.000	Harm
Qwen-plus	Economic	4.61	0.182	0.009	Ratios

Local models - Exp. 3a
(Ratio vs. harm)

Llama 3.2 (3.2B)	Moral	4.41	0.094	0.000	Ratios
Mistral (7.2B)	Moral	1.61	0.024	0.000	Harm
Mistral nemo (12.2B)	Moral	3.46	0.010	0.000	Harm
Llama 3.2 (3.2B)	Economic	3.50	0.043	0.123	
Mistral (7.2B)	Economic	1.51	0.021	0.000	Harm
Mistral nemo (12.2B)	Economic	3.48	0.010	0.125	

Humans - Exp. 3b
(Ratio vs. utility)

	Moral	1.28	0.066	0.000	Utility
Humans	Moral	3.91	0.101	0.000	Ratios
	Economic	1.23	0.037	0.000	Utility
Humans	Economic	3.28	0.099	0.033	Utility

Frontier models - Exp. 3b
(Ratio vs. utility)

Claude Sonnet 4.5	Moral	4.18	0.075	0.000	Ratios
Deepseek	Moral	1.00	0.000	0.000	Utility
Gemini 2.5 flash	Moral	2.53	0.169	0.000	Utility
GPT-4.1 mini	Moral	2.16	0.068	0.000	Utility
GPT-4o mini	Moral	3.12	0.059	0.000	Utility
Mistral Medium	Moral	3.69	0.028	0.000	Ratios
Qwen-plus	Moral	2.44	0.143	0.000	Utility
Claude Sonnet 4.5	Economic	2.47	0.128	0.000	Utility
Deepseek	Economic	1.00	0.000	0.000	Utility
Gemini 2.5 flash	Economic	2.40	0.144	0.000	Utility
GPT-4.1 mini	Economic	2.38	0.129	0.000	Utility
GPT-4o mini	Economic	2.44	0.041	0.000	Utility
Mistral Medium	Economic	3.11	0.024	0.000	Utility
Qwen-plus	Economic	2.16	0.139	0.000	Utility

Local models - Exp. 3b
(Ratio vs. utility)

Llama 3.2 (3.2B)	Moral	3.42	0.053	0.001	Utility
Mistral (7.2B)	Moral	1.52	0.015	0.000	Utility
Mistral nemo (12.2B)	Moral	3.06	0.013	0.000	Utility
Llama 3.2 (3.2B)	Economic	3.13	0.035	0.000	Utility
Mistral (7.2B)	Economic	1.56	0.015	0.000	Utility

Table 6: Detailed results from Experiment 3. Mean ratings (M) with standard errors (SE) are shown on a 1-6 scale, where values greater than 3.5 indicate preference for the high Weber ratio option and values smaller 3.5 indicate preference for the alternative (low harm in Experiment 3a, high utility in Experiment 3b). The p-value tests whether the mean differs from 3.5 (no preference). The 'favoring' column indicates which option was significantly preferred. *(continued)*

model	condition	M	SE	p	favoring
Mistral nemo (12.2B)	Economic	3.01	0.013	0.000	Utility

Table 7: Summary of model preferences in Experiment 3. Numbers indicate how many models (out of 10 frontier models or 3 local models) significantly preferred each option. 'Neither' indicates no significant preference ($p > 0.05$).

condition	Number of models favoring											
	<i>Humans</i>				<i>Frontier models</i>				<i>Local models</i>			
	Ratios	Utility	Harm	Neither	Ratios	Utility	Harm	Neither	Ratios	Utility	Harm	Neither
Exp. 3a (Ratio vs. harm)												
Moral	2		0	0	2		3	2	1		2	0
Economic	2		0	0	4		2	1	0		1	2
Exp. 3b (Ratio vs. utility)												
Moral	1	1		0	2	5		0	0	3		0
Economic	0	2		0	0	7		0	0	3		0

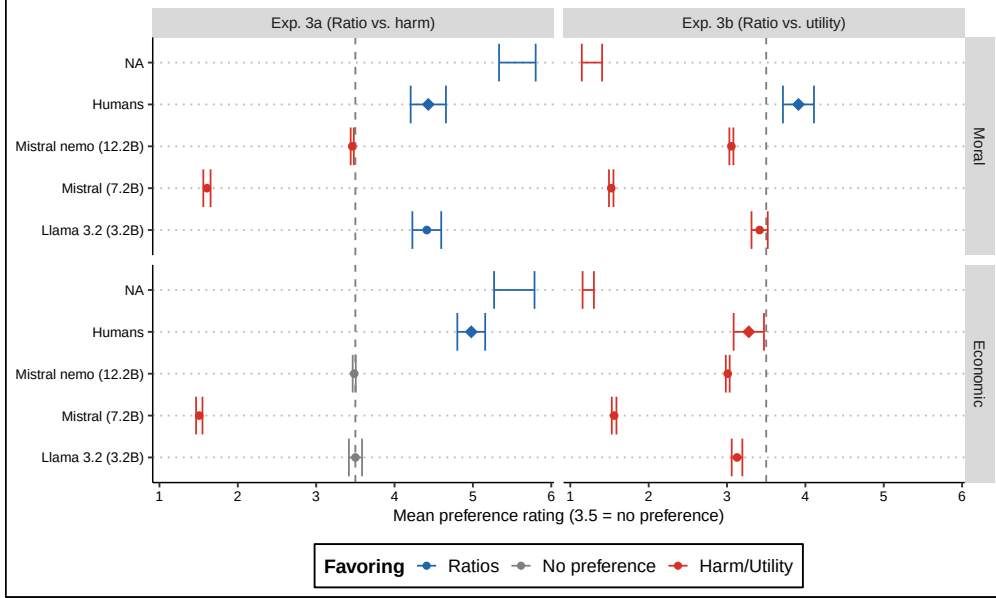


Figure 7: Forest plot of preferences between high-ratio and alternative options in Experiment 3 for local models. Points show mean preference ratings with 95% confidence intervals. The vertical dashed line at 3.5 indicates no preference; values above favour the high Weber ratio option, values below favour the alternative (low harm in Exp. 3a, high utility in Exp. 3b). Humans are shown as larger diamonds. Colour indicates the significantly favoured option (blue = ratio, red = alternative, grey = no significant preference).

1.4.2.2 Local models When considering (smaller) local LLMs, one model preferred Weber ratios over harm in moral situations, while two preferred harm over Weber ratios. In economic situations, one model preferred harm over Weber ratios, while two had no preference. When pitting ratios against utility, all models prioritized utility in both the moral and economic situations.

1.4.2.3 Summary Experiment 3 simulations showed some qualitative differences between human and LLM behavior.

Many of the LLMs were much more sensitive to harm than humans. While humans prioritized Weber ratios over harm, half of the frontier LLMs prioritized harm in moral situations, while a third prioritized harm in economic situations.

The LLMs also failed to replicate the preference for Weber ratios over utility in moral situations, with all but 2 models prioritizing utility, which also showed no preference for Weber ratios over harm. However, like humans, the LLMs prioritized utility in economic situations.

1.4.3 Experiment 6

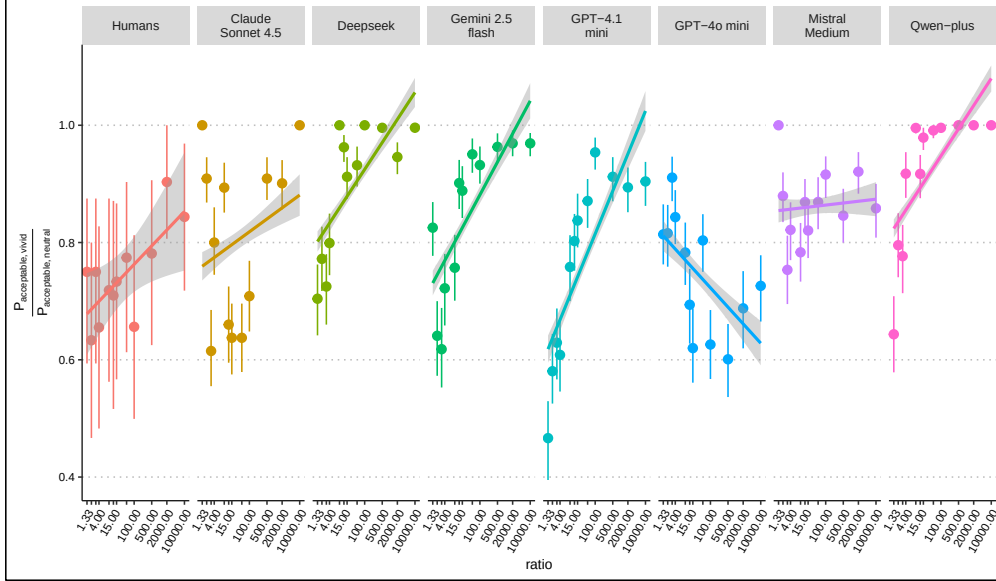


Figure 8: Risk ratio of acceptability judgements as a function of beneficiary:victim Weber ratio in Experiment 6 for frontier models. The risk ratio compares acceptability rates when victims’ deaths were portrayed vividly or in the same neutral language as the beneficiaries’ deaths, $RR = \frac{P_{\text{acceptable, vivid}}}{P_{\text{acceptable, neutral}}}$. Points show mean risk ratios with 95% confidence intervals. Regression lines show the relationship between Weber ratio and risk ratio. In humans, higher Weber ratios are associated with higher risk ratios, indicating that the effect of the salience of the victims’ deaths portrayals is reduced for higher beneficiary:victim ratios. Seven frontier models reproduced this positive correlation, one showed a negative correlation (GPT-4o mini), and two showed no correlation (Claude Sonnet 4.5, Mistral Medium).

1.4.3.1 Frontier models Experiment 6 assessed how acceptability ratings depend on Weber ratios when the deaths of the victims and the beneficiaries are equally salient, and when the deaths of the victims are made more vivid and salient. Our critical measure is the risk ratio comparing acceptability rates when the deaths of the victims are vividly portrayed versus when they are presented with the same neutral language as the beneficiaries, $RR = \frac{P_{\text{acceptable, vivid}}}{P_{\text{acceptable, neutral}}}$. In humans, this risk ratio is positively correlated with the beneficiary:victim Weber ratio, suggesting that the effect of victim salience decreases for higher Weber ratios.

The results for the frontier models are shown in Figure 8 and Table 8. Seven models showed the same positive correlation as humans, one showed a negative correlation (GPT-4o mini), and two showed no correlation (Claude-Sonnet-4.5 and Mistral Medium).

Table 8: Correlations between Weber ratio and risk ratio in Experiment 6. The risk ratio $RR = \frac{P_{\text{acceptable, vivid}}}{P_{\text{acceptable, neutral}}}$ measures how much the saliency of the victims’ death reduces acceptability judgments. Positive correlations indicate that this effect diminishes at higher Weber ratios (matching human behavior). Correlation coefficients (ρ), p-values (p), and significance levels (sig: * $p < .05$, ** $p < .01$, *** $p < .001$, ns = not significant) are shown. Most frontier models showed significant positive correlations, while local models showed negative or no correlations.

model	ρ	p	sig
Frontier models			
Humans	0.110	0.041	*
Claude Sonnet 4.5	0.018	0.370	ns
Deepseek	0.290	<0.001	***
Gemini 2.5 flash	0.280	<0.001	***
GPT-4.1 mini	0.330	<0.001	***
GPT-4o mini	-0.150	<0.001	***
Mistral Medium	0.005	0.786	ns
Qwen-plus	0.340	<0.001	***
Local models			
Humans	0.110	0.041	*
Llama 3.2 (3.2B)	-0.036	0.064	.
Mistral (7.2B)	-0.040	0.038	*
Mistral nemo (12.2B)	-0.012	0.525	ns

We also present the acceptability ratings and model fits in Figure 9 and Table 9, respectively.

Table 9: Psychophysical model parameter estimates for Experiment 6. The w parameter reflects precision in the representations of the numbers of victims and beneficiaries. Higher values indicate greater uncertainty; values greater than 2 indicate the model hit the upper bound of the estimation procedure. The α parameter reflects the attentional weight of the victims. Empty cells indicate convergence failure. Most models showed higher α values in the vivid than neutral condition.

Model	Condition	w		α	
		M	SE	M	SE
Humans					
Humans	Neutral	0.820	0.056	1.03	0.094
Humans	Vivid	1.154	0.090	1.32	0.141
	Neutral	0.162	0.144	1.00	0.266
	Vivid	0.511	0.028	1.00	0.040
Frontier models					
Claude Sonnet 4.5	Neutral	0.943	0.130	1.88	0.300
Claude Sonnet 4.5	Vivid	1.114	0.137	1.95	0.310
Deepseek	Neutral	0.886	0.061	1.00	0.102
Deepseek	Vivid	0.829	0.049	1.68	0.114
Gemini 2.5 flash	Neutral	0.695	0.034	1.06	0.055
Gemini 2.5 flash	Vivid	0.717	0.070	1.31	0.108
GPT-4.1 mini	Neutral	0.485	0.127	1.00	0.178
GPT-4.1 mini	Vivid	0.907	0.056	1.00	0.094
GPT-4o mini	Neutral	1.503	0.113	1.00	0.157
GPT-4o mini	Vivid	2.000	0.228	1.00	0.264
Mistral Medium	Neutral	0.910	0.071	1.87	0.164
Mistral Medium	Vivid	1.139	0.082	2.38	0.266
Qwen-plus	Neutral	0.798	0.080	1.00	0.136
Qwen-plus	Vivid	0.760	0.037	1.00	0.062
Local models					
Llama 3.2 (3.2B)	Neutral	1.190	0.129	1.00	0.202

Table 9: Psychophysical model parameter estimates for Experiment 6. The w parameter reflects precision in the representations of the numbers of victims and beneficiaries. Higher values indicate greater uncertainty; values greater than 2 indicate the model hit the upper bound of the estimation procedure. The α parameter reflects the attentional weight of the victims. Empty cells indicate convergence failure. Most models showed higher α values in the vivid than neutral condition. (*continued*)

Model	Condition	M	SE	M	SE
Llama 3.2 (3.2B)	Vivid	2.000	0.212	1.00	0.245
Mistral (7.2B)	Neutral	1.795	0.231	1.00	0.288
Mistral (7.2B)	Vivid	2.000	0.479	1.29	0.575
Mistral nemo (12.2B)	Neutral	2.000	0.641	1.57	1.066
Mistral nemo (12.2B)	Vivid	2.000	0.714	1.97	1.237

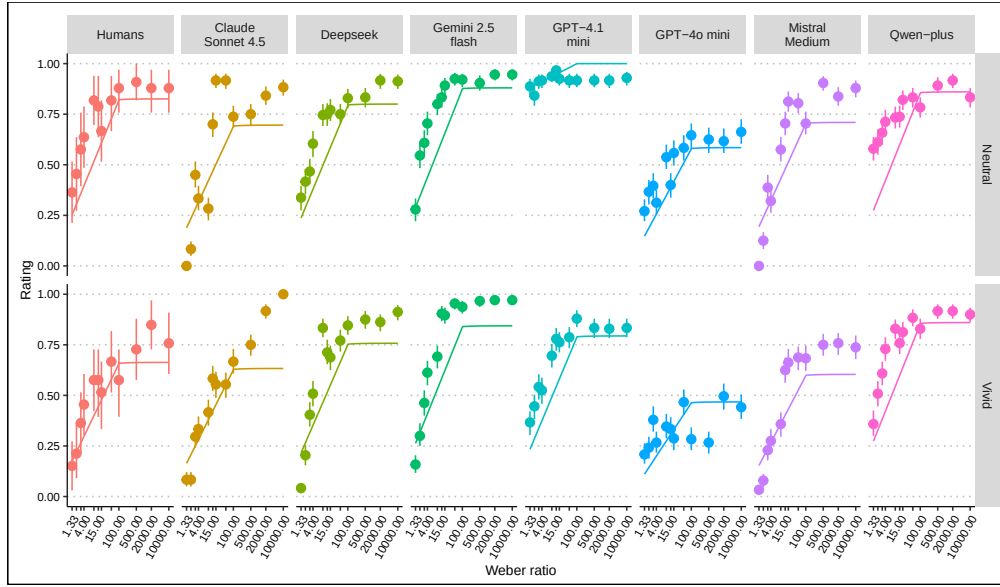


Figure 9: Acceptability of sacrificial decisions as a function of beneficiary:victim Weber ratio in Experiment 6 for frontier models. Data show binarized responses for neutral (top) and vivid (bottom) portrayals of the victims' deaths. Points represent mean acceptability with 95% confidence intervals. Curves show psychophysical model fits.

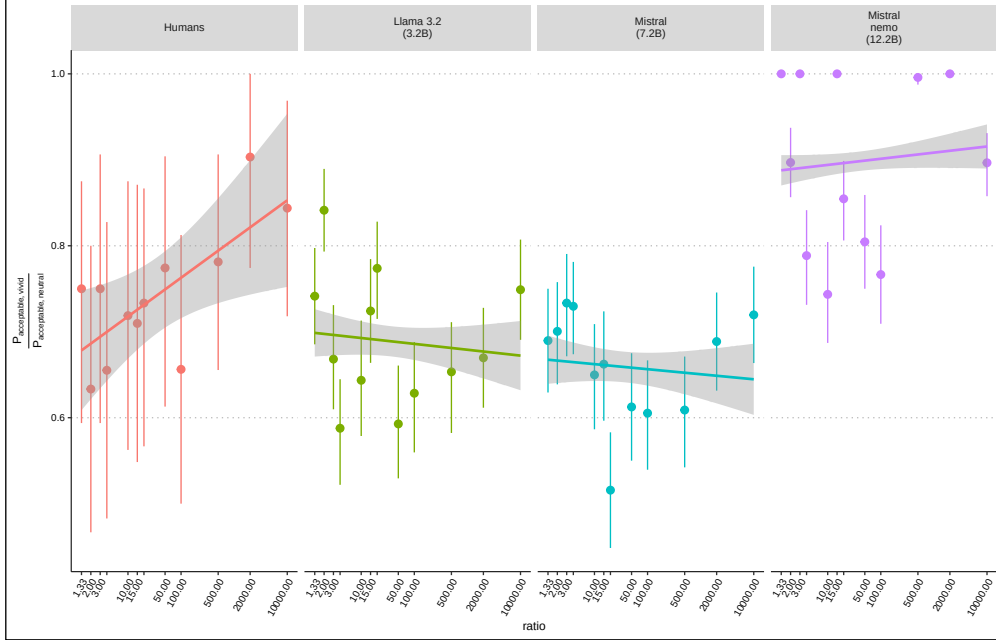


Figure 10: Risk ratio of acceptability judgements as a function of beneficiary:victim Weber ratio in Experiment 6 for local models. The risk ratio compares acceptability rates when victims’ deaths were portrayed vividly or in the same neutral language as the beneficiaries’ deaths, $RR = \frac{P_{\text{acceptable, vivid}}}{P_{\text{acceptable, neutral}}}$. Points show mean risk ratios with 95% confidence intervals. Regression lines show the relationship between Weber ratio and risk ratio. In humans, higher Weber ratios are associated with higher risk ratios, indicating that the effect of the salience of the victims’ deaths portrayals is reduced for higher beneficiary:victim ratios. Local models largely failed to reproduce the human pattern: one model (Mistral 7.2B) showed a negative correlation, while the other two models showed no correlation.

1.4.3.2 Local models We performed similar analyses for the local models. While the Mistral (7.2B parameters) showed a negative correlation between the risk ratio and the Weber ratio, the other two models showed no correlation.

We also present the acceptability ratings and model fits in Figure 11 and Table 9, respectively.

1.4.3.3 Summary Overall, these results suggest that the models generally reproduced the correlation between the risk ratio and the Weber ratio, though the quantitative fit is of variable quality.

1.5 Overall Summary

The frontier LLMs showed several key similarities to human moral judgment. Most models demonstrated sensitivity to beneficiary:victim Weber ratios, greater acceptance of sacrifices in economic versus moral decisions, and a decreasing effect of victim salience at higher Weber ratios. However, the quantitative fit to the psychophysical model varied considerably across models, and some models failed to show expected patterns.

These similarities appeared to depend on model size and presumably on massive amounts of training data, as the smaller local models largely failed to reproduce human behavior. Local models showed no discernible sensitivity to Weber ratios in Experiment 2 and exhibited inconsistent patterns across other experiments.

However, all models behaved qualitatively differently from humans in Experiment 3. They were more sensitive to both harm and utility than humans, especially in moral situations. In particular, no model showed a preference for Weber ratios over both harm and utility in moral situations.

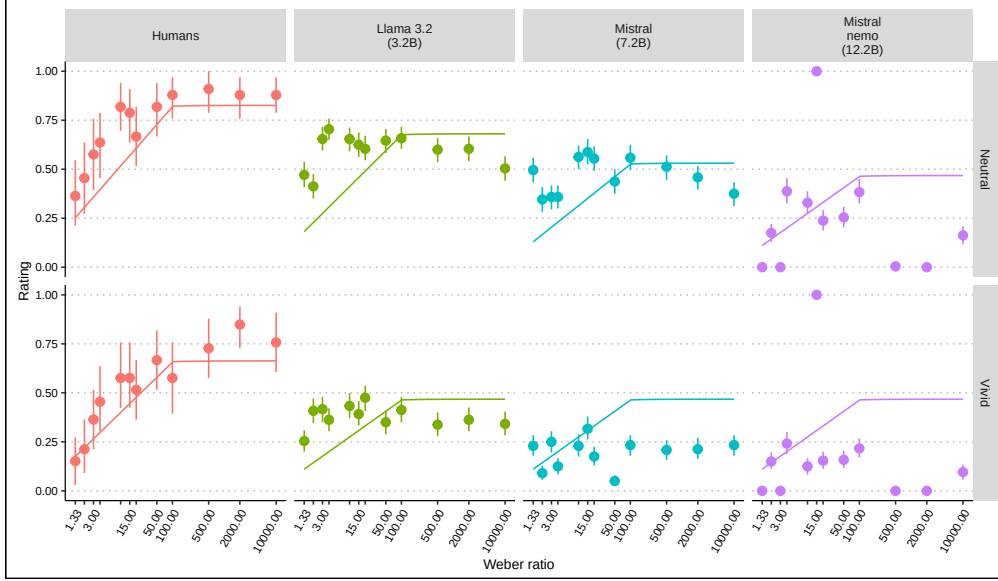


Figure 11: Acceptability of sacrificial decisions as a function of beneficiary:victim Weber ratio in Experiment 6 for local models. Data show binarized responses for neutral (top) and vivid (bottom) portrayals of the victims’ deaths. Points represent mean acceptability with 95% confidence intervals. Curves show psychophysical model fits.

1.6 Summary to be included in main text

In 1, we started addressing the question of whether the human behavior can be learned from data alone (i.e., from positive evidence in learnability theory terminology). Specifically, we asked whether large language models (LLMs) could reproduce the human behavior in Experiments 2, 3 and 6. We tested several frontier (state-of-the-art) models (GPT-4o mini, GPT-4.1, GPT-4.1 mini, GPT-5 mini (minimal reasoning), GPT-5 mini (medium reasoning), GPT-5 mini (high reasoning), Claude Sonnet 4.5, Gemini 2.5 flash, Gemini 2.5 pro, Mistral Medium, Deepseek, and Qwen-plus) from a range of jurisdictions (China, France and USA) with potential differences in training corpora and alignment procedures (i.e., post-training procedures used to shape model behavior according to values determined by the model creators and their jurisdictions). We also used different versions of OpenAI’s GPT models because this allowed us to set the reasoning effort parameter in more recent versions of the model (GPT-5 mini; this parameter controls how many internal and intermediate reasoning steps the model generates before producing its final response). We also explored several smaller models run on a local computer to test whether the massive scale for frontier models would be required to reproduce human results, namely Llama 3.2 (3.2B), Mistral (7.2B), and Mistral nemo (12.2B). Throughout these simulations, we kept the procedure as close as possible to the experimental procedures (see 1 for details).

The frontier LLMs showed several key similarities to human moral judgment. Most models demonstrated sensitivity to beneficiary:victim Weber ratios (Experiments 2 and 6), greater acceptance of sacrifices in economic versus moral decisions (Experiment 2), and a decreasing effect of victim salience at higher Weber ratios (Experiment 6). However, the quantitative fit to the psychophysical model varied considerably across models, and some models failed to show expected patterns.

These similarities appeared to depend on model size and presumably on massive amounts of training data, as the smaller local models largely failed to reproduce human behavior and showed no discernible sensitivity to Weber ratios. While a sensitivity to Weber ratios appears to be learnable from data, the scale required might thus be implausible for human learners.

Critically, all models behaved qualitatively differently from humans in the moral conditions of Experiment 3. They were more sensitive to both harm and utility, especially in moral situations, and no model showed

a preference for Weber ratios over both harm and utility in moral situations.